Joohyung Lee

Website LinkedIn Scholar GitHub

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Research Summary

- My research asks what regularities modern learning systems, including foundation and world models, should respect, and how to enforce them at scale without sacrificing utility. I develop **structure-respecting representation learning**: translating analytic priors (symmetry, acquisition physics, temporal dynamics) into scalable self-supervised objectives and layer-local regularizers for **robust generalization under shift**. PhD direction: **complement** analytic symmetries with learned spatiotemporal symmetries and semantic/physical plausibility constraints.
- **Research keywords:** self-supervised representation learning; geometric/equivariant deep learning; robustness & distribution shift; spatiotemporal representation learning; constraint-based learning for generative and world models.

Education

Texas A&M University — College Station, TX, USA
M.S. in Electrical Engineering — *Aug 2014*

The University of Texas at Dallas — Richardson, TX, USA
B.S. in Electrical Engineering — *Aug 2012*

Research & Industry Experience

AITRICS — Seoul, South Korea
ML Researcher — *May 2022 – Present*

- **Geometric / structure-respecting self-supervised learning:** proposed *Soft Equivariance Regularization (SER)*, a plug-in regularizer that decouples **invariance at the final embedding** from **equivariance constraints on intermediate spatial feature maps**; improved ImageNet-1k linear-eval Top-1 by +0.84 with negligible overhead (*ICLR 2026* (accepted); first author) [1].
- **Temporal regularities for multimodal clinical forecasting:** proposed predictive-coding self-supervision for irregular, partially observed EHR trajectories (*ICLR 2023* TML4H Workshop; Best Paper Honorable Mention; corresponding author) and introduced a shared time-embedding with modality-aware attention to align asynchronous modalities (*MLHC 2023* Spotlight; co-first author; corresponding author) [3, 7].
- **Acquisition-induced nuisance variation:** introduced a lightweight debiasing objective for weakly supervised learning on whole-slide images, suppressing acquisition-induced nuisance variation (*ICASSP 2024*; first author) [2].

Korea Electronics Technology Institute (KETI) — Seongnam-si, South Korea
Senior Researcher — *Dec 2020 – May 2022*

- Developed ML pipelines for on-device face recognition and 3D point-cloud perception for autonomous-driving systems.

KAIST, Robotics & Computer Vision Lab — Daejeon, South Korea
Researcher — *Mar 2020 – Dec 2020*

- Proposed late-fusion 3D model respecting anisotropic imaging geometry of MRI (*MICCAI 2022*, first author) [4].

National Cancer Center — Gyeonggi-do, South Korea
Visiting Researcher (part-time, compensated appointment; concurrent with AITRICS) — *Aug 2022 – Dec 2025*
Researcher — *Oct 2016 – Mar 2020*

- Studied robustness and uncertainty of deep segmentation under limited supervision; introduced a variance-reduction strategy for rectal-cancer segmentation (*IEEE Access* 2019) [8].

Laboratory for Optical Diagnosis & Imaging, Texas A&M University — College Station, TX, USA
Student Researcher — *Mar 2013 – Jun 2014*

- Developed ML for *in vivo* fluorescence lifetime imaging for oral-cancer detection (*Photochem. Photobiol.*; co-first) [10].

Teaching & Mentoring

Texas A&M University, Department of Electrical Engineering — College Station, TX, USA
Grader (Signals and Systems) — *Jan 2014 – Apr 2014*

- Created grading rubrics and evaluated coding assignments, written homework, and exams.

GEM Center, The University of Texas at Dallas — Richardson, TX, USA
Math Tutor — *Jan 2011 – May 2011*

- Provided one-on-one tutoring in undergraduate mathematics and physics.

Publications & Preprints

Selected Conference Publications

[1] **Soft Equivariance Regularization for Invariant Self-Supervised Learning.**
Joohyung Lee, Changhun Kim, Hyunsu Kim, Kwanhyung Lee, Juho Lee.
ICLR 2026 (accepted). [pdf]

[2] **Compact and De-Biased Negative Instance Embedding for Multi-Instance Learning on Whole-Slide Image Classification.**
Joohyung Lee, Heejeong Nam, Kwanhyung Lee, Sangchul Hahn.
ICASSP 2024. [pdf]

[3] **Learning Missing Modal Electronic Health Records with Unified Multi-modal Data Embedding and Modality-Aware Attention.**
Kwanhyung Lee, Soojeong Lee, Sangchul Hahn, Heejung Hyun, Edward Choi, Byungeun Ahn, **Joohyung Lee**.
MLHC 2023. **Spotlight**. (**Co-first author**; **Corresponding author**). [pdf]

[4] **Moving from 2D to 3D: Volumetric medical image classification for rectal cancer staging.**
Joohyung Lee, Jieun Oh, Inkyu Shin, You-sung Kim, Dae Kyung Sohn, Tae-sung Kim, In So Kweon.
MICCAI 2022. [pdf] [video]

Workshop

[5] **Structure-Aware Set Transformers: Temporal and Variable-Type Attention Biases for Asynchronous Clinical Time Series.**
Joohyung Lee, Kwanhyung Lee, Changhun Kim, Eunho Yang.
ICLR 2026 Workshop on TSALM (accepted). [pdf]

[6] **Status-Aware Self-Supervised Forecasting for Irregular Clinical Time Series.**
Kwanhyung Lee, **Joohyung Lee**, Jongheon Kim, Hanhsang Chul, Eunho Yang.
ICLR 2026 Workshop on TSALM (accepted; **co-first author**). [pdf]

[7] **Self-Supervised Predictive Coding with Multimodal Fusion for Patient Deterioration Prediction in Fine-grained Time Resolution.**
Kwanhyung Lee, John Won, Heejung Hyun, Sangchul Hahn, Edward Choi, **Joohyung Lee**.
ICLR 2023 Workshop on Trustworthy ML for Healthcare. Oral presentation; **Best Paper Honorable Mention**. (**Corresponding author**). [pdf]

Journal

[8] **Reducing the Model Variance of Rectal Cancer Segmentation Network.**
Joohyung Lee, Ji Eun Oh, Min Ju Kim, Bo Yun Hur, Dae Kyung Sohn.
IEEE Access, 2019. [pdf]

[9] **Tumor Identification in Colorectal Histology Images Using a Convolutional Neural Network.**
Hongjun Yoon, **Joohyung Lee**, Ji Eun Oh, Hong Rae Kim, Seonhye Lee, Hee Jin Chang, Dae Kyung Sohn.
Journal of Digital Imaging, 2018. [pdf]

[10] **Objective Detection of Oral Carcinoma with Multispectral Fluorescence Lifetime Imaging *In Vivo*.**
Bilal H Malik, **Joohyung Lee**, Shuna Cheng, Rodrigo Cuenca, Joey M Jabbour, Yi-Shing Lisa Cheng, John M Wright, Beena Ahmed, Kristen C Maitland, Javier A Jo.
Photochemistry and Photobiology, 2016. (**Co-first author**). [pdf]

In Preparation

[11] **Decoupling Equivariance from Invariant Self-Supervised Learning via Null-Space Projection.**
Joohyung Lee, Changhun Kim, Kwanhyung Lee.
Manuscript in preparation.

Honors & Recognition

- **MLHC 2023**: Spotlight paper.
- **ICLR 2023 TML4H Workshop**: Best Paper Honorable Mention; Oral presentation.

References

Prof. Eunho Yang (Research supervisor)
Associate Professor, Kim Jaechul Graduate School of AI, KAIST; Email: eunhoy@kaist.ac.kr

Prof. Juho Lee (Research collaborator)
Associate Professor, Kim Jaechul Graduate School of AI, KAIST; Email: juholee@kaist.ac.kr

Prof. Dae Kyung Sohn (Research supervisor)
Professor, Dept of Public Health & Artificial Intelligence, National Cancer Center; Email: gsgsbal@ncc.re.kr